INDEPENDENT INVESTIGATOR REFERENCE · MICROSOFT CLOUD AUDIT LOGS GUIDE

Microsoft Cloud Forensic Evidence Map

Audit, Diagnostic, Security, and Operational Evidence for Microsoft Cloud Investigations

What evidence source should an investigator examine, whether it should exist, and when is it relevant?

18 evidence sources · 13 Microsoft products · 154 official Microsoft references
Checked against Microsoft Learn on 2026-10-13

No evidence does not automatically mean no activity occurred.

Before drawing a conclusion, establish:

1. Was the evidence generated?
2. Did collection or auditing require configuration?
3. Was it configured at the time of the incident?
4. Where was the evidence routed?
5. What retention applied?
6. Does the evidence still exist?
7. Did the investigator have sufficient permissions to retrieve it?

AVAILABILITY LEGEND

AUTO	Evidence is generated or collected automatically
CONFIG	Auditing, collection, telemetry, or a feature must be configured
EXPORT	Native evidence can be exported or routed
LICENSE	Availability depends on licensing
DEPLOY	Product deployment or onboarding is required
POLICY	Evidence detail depends on an audit or logging policy
ON DEMAND	Evidence is produced only when requested
SUPPORTING	Contextual evidence rather than a primary audit trail

VERIFICATION STATUS

✓ Verified	Every field checked against current Microsoft documentation
! Conditional	Usable, with listed gaps still to confirm
? Verification required	Not yet confirmed; do not rely on it
✕ Deprecated or replaced	Retiring or replaced; use the named successor

TERMINOLOGY

- Azure Activity Log is control-plane evidence. It doesn't record every operation inside a resource.
- Azure resources generate Resource Logs, but Azure Monitor does not collect them by default. A Diagnostic Setting or another documented collection configuration must exist for historical collection.
- Diagnostic settings are a collection and routing configuration, not a log.
- AzureDiagnostics is not the universal table; most services write to resource-specific tables.
- Microsoft Sentinel correlates evidence routed to it; it doesn't produce the underlying evidence.
- Defender evidence is security telemetry, alerts, and incidents, not an audit log.

RESOURCE LOG CATEGORIES

Categories and tables differ for every Azure service and are not printed here. See Supported Resource log categories for Azure Monitor and the interactive map.

NOT YET CATALOGUED

These planes have no verified catalog entries yet and are intentionally left off this edition:

- Enhanced Windows Telemetry
- Containers and Applications
- Power Platform and Dynamics 365
- Microsoft Intune

EVIDENCE PLANE	MICROSOFT PRODUCT	LOG OR EVIDENCE SOURCE	AVAILABILITY	WHERE TO INVESTIGATE	PRESERVATION OR DESTINATION	INVESTIGATION RELEVANCE	MICROSOFT LEARN
01 IDENTITY	**Microsoft Entra ID** AUDIT	**Audit logs** ✓ Verified	AUTO LICENSE EXPORT Automatic with limited native retention Native retention: 7 days (Microsoft Entra ID Free); 30 days (Microsoft Entra ID P1 or P2)	Microsoft Entra admin center; Microsoft Graph audit logs API; Log Analytics (AuditLogs) when routed by a diagnostic setting.	• Microsoft Entra diagnostic setting to a Log Analytics workspace • Microsoft Entra diagnostic setting to an Azure Storage account (archive) • Microsoft Entra diagnostic setting to Event Hubs • Microsoft Purview Audit (Premium) retention for Microsoft Entra audit records (E5-class or Purview Suite licensing) • Download from the portal as CSV or JSON	• Who granted a user membership in an administrative group, and when • What changes were made to an application or service principal • How many password resets were made in a period • What operations a specific agent identity performed	Learn: What is Microsoft Entra? Evidence docs: Audit logs in Microsoft Entra ID
	Microsoft Entra ID SIGN-IN	**Sign-in logs** ! Conditional · 1 open gap	AUTO LICENSE EXPORT Automatic with limited native retention Native retention: 7 days (Microsoft Entra ID Free); 30 days (Microsoft Entra ID P1 or P2)	Microsoft Entra admin center; Microsoft Graph /auditLogs/signIns; Log Analytics sign-in tables when routed.	• Microsoft Entra diagnostic setting to a Log Analytics workspace (select each sign-in category) • Microsoft Entra diagnostic setting to an Azure Storage account (archive) • Microsoft Entra diagnostic setting to Event Hubs	• Sign-in pattern of a user: when, from where, with which client and application • Whether sign-ins succeeded or failed, and which Conditional Access policies applied • Non-interactive sign-ins made by clients on behalf of a user • Service principal and managed identity authentication activity	Learn: What is Microsoft Entra? Evidence docs: What are Microsoft Entra sign-in logs?
02 AZURE CONTROL PLANE	**Azure Monitor** CONTROL PLANE	**Azure Activity Log** ! Conditional · 1 open gap	AUTO EXPORT Automatic with limited native retention Native retention: 90 days	Azure portal Activity log (Monitor, subscription, resource group, or resource); Activity Log REST API; Log Analytics (AzureActivity) when exported.	• Diagnostic setting to a Log Analytics workspace (AzureActivity table; retention up to 12 years) • Diagnostic setting to an Azure Storage account (insights-activity-logs container, hourly PT1H.json blobs) • Diagnostic setting to Event Hubs • Download from the portal as CSV	• Who created, changed, or deleted an Azure resource, and when • Changes made through Azure Resource Manager, such as a key vault access policy change • Which identity initiated a management operation (caller, claims, authorization, caller IP address) • Deployment errors and operations that share a correlation ID • Role assignment and custom role definition changes, recorded in the Administrative category	Learn: Azure Monitor overview Evidence docs: Activity log in Azure Monitor
03 AZURE RESOURCE AND DATA PLANE	**Azure Monitor** RESOURCE/DATA PLANE	**Azure Resource Logs** ✓ Verified	CONFIG EXPORT Configuration required Native retention: None: not collected or stored by Azure Monitor unless a diagnostic setting routes them	The destination chosen in the resource's diagnostic setting: Log Analytics workspace, Azure Storage account, Event Hubs, or a partner integration.	• Log Analytics workspace (table retention up to 12 years total) • Azure Storage account (insights-logs-{category} containers, hourly PT1H.json blobs); an immutability policy prevents modification • Event Hubs • Azure Monitor partner integrations	• Data-plane activity inside a resource that the Activity Log does not record • Request-level detail (caller, operation, result) for resource types whose schema includes it	Learn: Azure Monitor overview Evidence docs: Resource logs in Azure Monitor
	Azure Key Vault RESOURCE/DATA PLANE	**Key Vault resource logs (AuditEvent)** ! Conditional · 1 open gap	CONFIG EXPORT Configuration required Native retention: None: not collected or stored until a diagnostic setting routes them	The destination of the vault's diagnostic setting: Log Analytics workspace, Azure Storage account, or Event Hubs.	• Diagnostic setting to a Log Analytics workspace • Diagnostic setting to an Azure Storage account • Diagnostic setting to Event Hubs	• Which identity accessed or tried to access a secret, key, or certificate, and when • Changes to the vault, its access policies, keys, and secrets • Authentication API requests made to the vault	Learn: About Azure Key Vault Evidence docs: Azure Key Vault logging
	Azure Storage RESOURCE/DATA PLANE	**Storage resource logs (StorageRead, StorageWrite, StorageDelete)** ✓ Verified	CONFIG EXPORT Configuration required Native retention: None: not collected or stored until a diagnostic setting routes them	The destination of each service's diagnostic setting: Log Analytics workspace, Azure Storage account, or Event Hubs.	• Diagnostic setting to a Log Analytics workspace • Diagnostic setting to a separate Azure Storage account • Diagnostic setting to Event Hubs	• Which principal or token read, wrote, or deleted data, and when • How requests were authorized, including unexpected Shared Key, SAS, or anonymous access • Failed requests: authorization, throttling, network, and time-out errors	Learn: Introduction to Azure Storage Evidence docs: Monitor Azure Blob Storage
04 AZURE NETWORK AND REMOTE ACCESS	**Azure Firewall** NETWORK	**Azure Firewall resource logs** ✓ Verified	CONFIG EXPORT Configuration required Native retention: None: not collected or stored until a diagnostic setting routes them	The destination of the firewall's diagnostic setting; Log Analytics resource-specific AZFW* tables.	• Diagnostic setting to a Log Analytics workspace (resource-specific mode recommended) • Diagnostic setting to an Azure Storage account • Diagnostic setting to Event Hubs	• Whether traffic between a source and destination was allowed or denied, and by which rule • Outbound application (FQDN) access through the firewall • Threat-intelligence and IDPS signature matches • DNS queries handled by the firewall's DNS proxy	Learn: What is Azure Firewall? Evidence docs: Monitor Azure Firewall
	Azure Network Watcher NETWORK	**Virtual network flow logs** ! Conditional · 2 open gaps	CONFIG EXPORT Configuration required Native retention: Set per flow log in days on Standard general-purpose v2 storage accounts; 0 keeps data until manually deleted	Azure Storage account (raw flow logs); Log Analytics NTANetAnalytics (traffic analytics output).	• Storage account retention set on the flow log (Standard general-purpose v2 accounts) • Traffic analytics to a Log Analytics workspace	• Whether a connection between two IP addresses and ports occurred, and whether it was allowed or denied • Bytes and packets per flow, including stateless flows • Traffic to private endpoints, recorded at the source VM • Which network security group or security admin rule allowed or denied a flow	Learn: What is Azure Network Watcher? Evidence docs: Virtual network flow logs
05 WINDOWS AND COMPUTE	**PowerShell (Windows PowerShell 5.1 and PowerShell 7)** WINDOWS EVENT	**PowerShell Operational logs (script block and module logging)** ✓ Verified	CONFIG POLICY EXPORT Auditing policy dependent Native retention: Size-bound local log. Microsoft troubleshooting guidance for Azure VMs gives 15,728,640 bytes as the default maximum size of Microsoft-Windows-PowerShell/Operational.	On the device: Event Viewer > Applications and Services Logs.	• Export to EVTX with wevtutil epl • Azure Monitor Agent with a data collection rule to a Log Analytics workspace (Event table)	• Content of PowerShell scripts and commands run on a host, interactively or through automation (4104) • Cmdlet invocation and parameter binding for modules with logging enabled (4103)	Learn: PowerShell security features Evidence docs: about_Logging (Windows PowerShell 5.1)
	Windows (Windows Event Log) WINDOWS EVENT	**Security event log** ✓ Verified	POLICY EXPORT Auditing policy dependent Native retention: Size-bound local log. Microsoft troubleshooting guidance for Azure VMs gives 20,971,520 bytes as the default maximum size of the Security log.	On the device: Event Viewer (Security log), Get-WinEvent, or an exported EVTX file.	• Export to EVTX with wevtutil epl • Azure Monitor Agent with a data collection rule to a Log Analytics workspace (Event table) • Microsoft Sentinel 'Windows Security Events via AMA' connector (SecurityEvent table)	• Process execution and command lines on a host, when process creation auditing and command-line inclusion are enabled (4688; 4689 on termination) • Successful and failed logons (4624, 4625) and account lockouts (4740), according to the audit policy • Password change and reset attempts (4723, 4724) and user account creation or enablement (4720, 4722) • Security group creation and changes (4727, 4731, 4735, 4737, 4754, 4755) • Audit log cleared (1102) and advanced audit policy overwritten (4719)	Learn: Advanced audit policy Evidence docs: Advanced Audit Policy Configuration
07 MICROSOFT DEFENDER ANTIVIRUS	**Microsoft Defender Antivirus** SECURITY TELEMETRY	**Microsoft-Windows-Windows Defender/Operational event log** ✓ Verified	AUTO EXPORT Automatic Native retention: Size-bound local log; default size not documented in reviewed sources	On the device: Event Viewer > Applications and Services Logs > Microsoft > Windows > Windows Defender > Operational.	• Export to EVTX with wevtutil epl • Azure Monitor Agent with a data collection rule to a Log Analytics workspace (Event table); select the Microsoft-Windows-Windows Defender/Operational channel with a custom filter • On onboarded devices, Defender for Endpoint telemetry (DeviceEvents in advanced hunting) and alerts	• Malware detections and the action taken: 1116 (detection) and 1117 (action), with threat name, path, detection origin and source, user, and process • Scans started, finished, or cancelled (1000, 1001, 1002) • Defender Antivirus configuration changes (5007) and scanning disabled (5010) • Detections that were allowed rather than remediated (1117 with an Allow action)	Learn: Microsoft Defender Antivirus overview Evidence docs: Review event logs and error codes to troubleshoot issues with Microsoft Defender Antivirus
08 MICROSOFT DEFENDER FOR ENDPOINT	**Microsoft Defender for Endpoint** SECURITY TELEMETRY	**Advanced hunting device tables (endpoint telemetry)** ✓ Verified	DEPLOY LICENSE EXPORT Deployment dependent Native retention: Advanced hunting returns data from the past 30 days	Microsoft Defender portal (advanced hunting); Microsoft Graph security API (runHuntingQuery).	• Onboard a Microsoft Sentinel workspace and configure analytics-tier retention for supported Defender XDR tables • Microsoft Defender Streaming API • Defender for Endpoint Raw Data Streaming API	• Process creation and command lines on onboarded devices (DeviceProcessEvents) • Network connections (DeviceNetworkEvents), file system events (DeviceFileEvents), and registry changes (DeviceRegistryEvents) • Sign-ins and other authentication events on devices (DeviceLogonEvents) and DLL loading (DeviceImageLoadEvents) • Security-control events such as Defender Antivirus and exploit protection (DeviceEvents) • Hunting across devices over the last 30 days of raw endpoint signals	Learn: Microsoft Defender for Endpoint overview Evidence docs: Proactively hunt for threats with advanced hunting in Microsoft Defender
	Microsoft Defender for Endpoint WINDOWS EVENT	**Microsoft-Windows-SENSE/Operational event log** ✓ Verified	DEPLOY SUPPORTING Deployment dependent Native retention: Size-bound local log; default size not documented in reviewed sources	On the device: Event Viewer > Applications and Services Logs > Microsoft > Windows > SENSE > Operational.	• Export to EVTX with wevtutil epl • Azure Monitor Agent with a data collection rule to a Log Analytics workspace (Event table)	• Whether the sensor was running and connected during a period: started (1), shut down (2), failed to start (3), contacted the server (4), failed to connect (5) • Onboarding failures, such as missing or unreadable onboarding parameters (6, 7)	Learn: Microsoft Defender for Endpoint overview Evidence docs: Review events and errors using Event Viewer
10 MICROSOFT 365 AND PURVIEW AUDIT	**Microsoft Purview Audit** AUDIT	**Unified audit log** ! Conditional · 1 open gap	AUTO LICENSE POLICY EXPORT Automatic Native retention: 180 days (Audit Standard); 1 year by default for Microsoft Entra ID, Exchange, OneDrive, and SharePoint records of Audit (Premium) users	Microsoft Purview portal (Audit search tool); Exchange Online PowerShell; Office 365 Management Activity API; Microsoft Graph Audit Search API.	• Audit (Premium) retention policies (up to 1 year; 10 years with the add-on) • Export search results to CSV • Office 365 Management Activity API to keep data longer than the default retention in another store	• Which user or admin performed an activity in Exchange Online, SharePoint, OneDrive, Teams, Power BI, or Microsoft Entra ID, and when • Activity by user, IP address, object, site, operation, or record type	Learn: Auditing solutions in Microsoft Purview Evidence docs: Search the audit log
13 MICROSOFT SECURITY AND DEFENDER XDR	**Microsoft Defender XDR** INCIDENT	**Incidents and alerts** ✓ Verified	LICENSE DEPLOY EXPORT Licensing dependent Native retention: Alerts are returned within the environment's retention policy; a fixed value is not documented in reviewed sources	Microsoft Defender portal (incidents and alerts); Microsoft Graph /security/incidents and /security/alerts_v2.	• Microsoft Defender XDR connector for Microsoft Sentinel (streams incidents, alerts, and advanced hunting events) • Microsoft Defender Streaming API • Export API results as JSON	• Which detections fired, on which devices, users, and mailboxes, and how they were grouped into an incident • Incident classification, determination, severity, status, and assignment	Learn: What is Microsoft Defender XDR? Evidence docs: List incidents
14 CORRELATION, COLLECTION, AND PRESERVATION	**Azure Monitor** COLLECTION CONFIGURATION	**Diagnostic Settings** ! Conditional · 1 open gap	CONFIG Configuration required Native retention: Not applicable (configuration object)	Azure portal (resource > Diagnostic settings); Az PowerShell; Microsoft Entra admin center for Microsoft Entra logs.		• Whether a resource's logs were being collected, which categories, and to which destination; establish this before concluding that missing logs mean no activity • Which collection mode (AzureDiagnostics or resource-specific) applies, and therefore which tables to query	Learn: Azure Monitor overview Evidence docs: Diagnostic settings in Azure Monitor
	Azure Monitor REPOSITORY	**Log Analytics workspace (Azure Monitor Logs)** ! Conditional · 1 open gap	CONFIG Configuration required Native retention: Analytics tables: 30 days by default (some log tables 90 days); analytics retention up to 2 years; total retention up to 12 years	Azure portal (Log Analytics); Az PowerShell; Microsoft Defender portal advanced hunting when the workspace is onboarded to Microsoft Sentinel.	• Analytics retention per table, up to 2 years • Total (long-term) retention per table, up to 12 years; retrieve older data with search jobs	• Correlating routed evidence from several sources with KQL in one place • Establishing which time range each table actually holds before drawing conclusions	Learn: Azure Monitor overview Evidence docs: Log Analytics workspace overview
	Microsoft Sentinel CORRELATION LAYER	**Microsoft Sentinel analytics, alerts, and incidents** ! Conditional · 1 open gap	DEPLOY CONFIG Deployment dependent Native retention: Governed by the underlying Log Analytics workspace and table settings	Microsoft Defender portal (Sentinel in the Defender portal); Azure portal until March 31, 2027; Microsoft Graph alerts_v2 filtered on serviceSource 'microsoftSentinel'.	• Workspace table retention (analytics and long-term) • Microsoft Sentinel data lake tier	• Which analytics rules fired and which entities they involved, as a pointer back to the original evidence records • Cross-source correlation over evidence routed into the workspace	Learn: What is Microsoft Sentinel? Evidence docs: What is Microsoft Sentinel security information and event management (SIEM)?

Each row is one catalog entry. Availability chips use text labels; color is supplementary.